Fanze Meng

Materials science, engineering mathematics, and computing

Japan | meng.f.0621@m.isct.ac.jp
dmpoolm.github.io | GitHub | LinkedIn

MSc student in computational materials at Institute of Science Tokyo, with degrees in materials science and engineering mathematics. Experience spans generative atomistic modelling, materials machine learning, experimental materials research, and software for industrial R&D.

Education

Institute of Science Tokyo Oct 2026 - present

MSc | Chemical Science / Computational Materials

Chalmers University of Technology Aug 2023 - Jun 2026

MSc | Engineering Mathematics and Computational Science

Queen Mary University of London Aug 2019 - Jul 2023

BEng | Materials Science and Engineering

Completed the Chalmers MSc in June 2026; began a second MSc focused on computational materials at Science Tokyo in October 2026.

Engineering experience

Volvo Cars | Materials & AI internship Feb 2025 - Jan 2026

- Developed and deployed Material Centre, an interactive materials-search application using Streamlit, Plotly, PCA, and distance metrics; containerised with Docker for internal user testing.
- Built CNN-Transformer models for stress-strain prediction from microstructure images of 90 representative volume elements; validation R^2 = 0.89, with Monte Carlo-dropout uncertainty estimates.
- Added a physics-informed variant with elastic-modulus, smoothness, and boundary-condition losses.

EDF, China R&D Center | R&D Assistant Jun 2024 - Sep 2024

Supported research on additive manufacturing for power-plant repair, analysing application cases and co-writing a technology roadmap. Researched HPC requirements for manufacturing, safety, data analysis, and materials R&D.

Master's thesis

Chalmers × Volvo Cars | Generative AI for robotics Oct 2025 - Jun 2026

- Co-developed a proof-of-concept LLM agent using the OpenAI GPT API and retrieval over ABB manuals to generate RAPID robot code.
- Implemented an HPC-based code-validation framework; benchmarked the workflow on approximately 50 manufacturing tasks and analysed scalability.

Current research

Institute of Science Tokyo | Tateyama-Ando Lab Oct 2026 - present

Physics-aware generative models for reconstructing periodic 3D atomic structures from pair distribution functions, targeting solid electrolytes for batteries.

- Implemented structure-to-distribution preprocessing, grouped dataset splits, and data-quality diagnostics.
- Implemented conditional flow-matching training and sampling, a diffusion baseline, and structural evaluation with short MLIP relaxation.

Current focus: data coverage, local bonding, and comparison with simpler baselines.

Research and experimental background

Tohoku University | COLABS research exchange Sep 2024 - Feb 2025

- Built a graph-similarity clustering pipeline for 3D electron diffraction candidates, reducing the number requiring manual verification by 97.2% on average across 7 datasets (2,127 structures).
- Replicated a catalyst-screening study with pretrained FAIR-Chem GNN models, including adsorbate-slab generation, ML relaxations, and comparison with literature adsorption energies.

Undergraduate laboratory research 2021 - 2023

Worked on thin-film fabrication for flexible displays and on aqueous flexible zinc-ion battery materials. Experience includes electrodeposition, electrode-separator fabrication, IR spectroscopy, XRD, SEM, and electrochemical testing.

Tools and interests

Programming and ML: Python, C, MATLAB/Simulink, Julia, SQL; PyTorch, TensorFlow, scikit-learn, XGBoost; generative models, uncertainty estimation, LLM agents, and RAG.

Scientific software: FAIR-Chem, ASE, RDKit; Linux, Git, Docker, Azure, Streamlit, Plotly; parallel computing, image analysis, and control systems.

Languages: Chinese (native), English (fluent; IELTS 7.5), Japanese (actively studying). **Personal interests:** hiking, cycling, driving.

Publications

Collaborative papers from undergraduate study.

B. Yang, F. Meng, J. Zhang, K. Chen, S. Meng, K. Cai, Y. Zhao, L. Dai. Engineered drug delivery nanosystems for tumor microenvironment normalization therapy. *Nano Today 49, 101766* (2023). doi:10.1016/j.nantod.2023.101766.

K. Gu, F. Meng. Former research and recent advances of metal-organic frameworks (MOF) for anti-cancer drug delivery. *Journal of Physics: Conference Series 2021, 012021* (2021). doi:10.1088/1742-6596/2021/1/012021.